

Fibrewise Characterization of Resolution Geometry

Fixed-Protocol Representation, Maximal Admissibility, and a Finite Uniqueness Attack

Matěj Rada

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Abstract

Resolution Geometry (RG) is not a protocol-free geometry selected by an experiment alone. Its primitive object is a declared admissible experiment $(\mathcal{E}, \Gamma)$, consisting of a complete statistical experiment and an operational contract. We identify the exact sense in which RG is canonical after this contract is fixed. Observational extensionality forces the equality-of-law quotient. In the regular finite classical module, sufficient-Markov invariance selects the Fisher–Rao metric up to normalization. A fixed comparison metric and threshold determine a unique hard spectral projector. We add the missing admissibility-completeness axiom: the final observable sector is the maximal resolved subobject satisfying every declared nuisance, stability, conditioning, coarse-graining, and transport gate. In finite-dimensional linear protocol fibres, this maximal subobject is the intersection of the corresponding admissible subspaces and is unique. Naturality and cocycle laws then determine transported copies and atlas gluing relative to the declared transition data.

These modules yield a fixed-protocol representation theorem: within the stated regular finite classical category, two observable geometries realizing the same $(\mathcal{E}, \Gamma)$ and satisfying quotient universality, the Fisher module, hard spectral resolution, admissibility soundness and completeness, natural transport, and composition are uniquely naturally isomorphic, up to the declared Fisher normalization. The theorem is grounded in established quotient universal properties, Čencov-type metric characterization, the spectral theorem and Borel functional calculus, lattice intersections of subspaces, and standard uniqueness of universal constructions. An exhaustive six-mode attack over all 4096 hard spectral/projected-sector pairs leaves exactly one survivor; soundness alone leaves a strict-subobject competitor, while completeness removes it. The result does not select Γ from $\mathcal{E}$, establish protocol-free uniqueness, or extend automatically to singular, infinite-dimensional, or quantum experiments. It identifies RG precisely as the canonical observable geometry inside each fixed admissibility-protocol fibre.

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1 Introduction

The foundational question is not whether every conceivable observation theory must equal Resolution Geometry. In an unrestricted class, no such uniqueness is plausible: an experiment does not determine a comparison scale, decision threshold, nuisance convention, stability class, or connection. The correct question is conditional:

$$(\mathcal{E}, \Gamma) \longmapsto \text{RG}(\mathcal{E}, \Gamma). \quad (1)$$

Here $\mathcal{E}$ declares the complete observation law and Γ declares the operational contract under which a direction counts as admissibly observable.

This shift is analogous to familiar geometric constructions. A manifold does not select a Riemannian metric, but a Riemannian manifold (M, g) determines its Levi–Civita connection from metric compatibility and vanishing torsion. Likewise, RG does not claim that $\mathcal{E}$ selects Γ . It claims that the bundled input $(\mathcal{E}, \Gamma)$ determines the observable quotient, distinguishability geometry, hard resolution, maximal admissible sector, and compatible transport under explicit hypotheses.

The earlier modular axioms already characterized the quotient, the regular classical metric module, the hard resolved projector, and natural gluing. The remaining finite-dimensional ambiguity was that admissibility had been stated only as containment,

$$W_{\text{obs|adm}} \subseteq \text{im } P_\tau \cap N^\perp \cap S \cap C \cap T, \quad (2)$$

which permits the zero sector and arbitrary strict subspaces. The finite attack reported here isolates the missing condition: admissibility must be complete as well as sound. Thus $W_{\text{obs|adm}}$ is the maximal resolved subobject satisfying all declared gates.

Contribution. The paper provides:

1. an exact definition of the fixed protocol fibre;
2. an admissibility soundness/completeness pair;
3. a proof of maximal-sector uniqueness in finite-dimensional linear fibres;

4. a modular fixed-protocol representation theorem;
5. a natural-isomorphism corollary across transported presentations;
6. an exhaustive finite counterexample search and axiom-independence witness;
7. explicit non-claims separating proved finite results from open extensions.

2 Established mathematical grounding

The result is an assembly theorem. Its load-bearing modules are established constructions used within explicitly stated domains.

2.1 Quotients and universal properties

For a complete statistical experiment $\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta})$, equality of laws defines

$$\theta \sim_{\mathcal{E}} \theta' \iff P_\theta = P_{\theta'}. \quad (3)$$

The quotient $Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}}$ has the standard factorization property for maps constant on equivalence classes. This is an ordinary quotient universal property; universal constructions are unique up to unique compatible isomorphism. The claim made here is no stronger than that standard fact at the quotient layer.

2.2 Fisher–Rao characterization

In the regular finite classical setting, a Čencov-type theorem characterizes the Fisher information metric, up to a positive scalar normalization, among smooth metric families invariant under sufficient Markov embeddings. Modern extensions establish related formulations for exponential families under explicit invariance hypotheses. RG imports this as a metric module; observational quotient logic alone does not select Fisher geometry.

2.3 Spectral functional calculus

For a positive comparison metric G and positive-semidefinite Fisher tensor F , the information operator

$$L = G^{-1}F \quad (4)$$

is self-adjoint in the G inner product. The finite-dimensional spectral theorem supplies a G -orthonormal eigenbasis, and Borel functional calculus supplies the canonical spectral projection

$$P_\tau = \mathbf{1}_{(\tau, \infty)}(L). \quad (5)$$

The projector is therefore not an extra coordinate after (F, G, τ) is fixed.

2.4 Maximal subobjects

The subspaces of a finite-dimensional vector space form a complete lattice: meets are intersections and joins are linear spans. Consequently, for a finite family of declared linear admissibility subspaces, their intersection is the unique greatest subspace contained in every member. This elementary lattice fact is the established grounding of admissibility completeness in the present theorem. In a more general category, the corresponding construction requires the relevant pullbacks or meets of subobjects; their existence is not assumed automatically outside the finite linear setting.

2.5 Blackwell order and information monotonicity

Blackwell comparison orders statistical experiments by informativeness: a less informative experiment is obtained by garbling a more informative one, equivalently under the classical decision-theoretic hypotheses it cannot improve every decision problem. RG uses only the compatible monotonicity requirement that parameter-independent processing cannot create distinguishability. Blackwell order does not choose G , τ , or the nuisance convention; it constrains admissible processing.

3 Declared admissible experiments

Definition 3.1 (Complete statistical experiment). *A complete statistical experiment is*

$$\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta}), \quad (6)$$

where Θ is a parameter or preparation domain, $\mathcal{O}$ is an outcome space, and P_θ is the complete outcome law at θ .

Definition 3.2 (Admissibility protocol). *An admissibility protocol is a tuple*

$$\Gamma = (G, \tau, N, S, C, T), \quad (7)$$

where G is a covariant positive comparison structure, τ is a resolution threshold or threshold family, N is a nuisance/gauge convention, S is a stability protocol, C is a conditioning and coarse-graining protocol, and T is transport/gluing data.

Definition 3.3 (Declared admissible experiment). *The primitive RG object is the bundled pair*

$$\mathbb{E} = (\mathcal{E}, \Gamma). \quad (8)$$

No component of Γ is inferred silently from inaccessible latent labels, undeclared priors, or an external physical interpretation.

Definition 3.4 (Protocol fibre). *Fix a protocol Γ . The protocol fibre Exp_Γ is the category whose objects are experiments carrying that protocol type and whose morphisms are experiment morphisms together with the prescribed covariant transport of all protocol data. The target fibre ObsGeom_Γ consists of quotient geometries, metric modules, resolved projectors, admissible sectors, and transition maps preserving the fixed contract.*

Remark 3.5. “Fixed protocol” means fixed as structured data, not fixed coordinate arrays. Under a change of representation, G , nuisance data, gates, and transports move covariantly. Fibrewise uniqueness is therefore uniqueness up to protocol-preserving natural isomorphism, not literal equality of matrices.

4 Axioms for fixed-protocol observable geometry

Axiom 4.1 (Observational extensionality). *Every experiment-internal construction Φ satisfies*

$$P_\theta = P_{\theta'} \implies \Phi(\theta) = \Phi(\theta'). \quad (9)$$

Axiom 4.2 (Naturality). *Observable constructions commute with experiment isomorphisms and with covariant transport of the complete protocol.*

Axiom 4.3 (Information order). *A parameter-independent garbling cannot increase the local distinguishability tensor.*

Axiom 4.4 (Off-discriminant regularity). *Continuous structures vary continuously and discrete invariants remain locally constant away from the declared discriminant, including threshold crossings, rank changes, nuisance-rank changes, conditioning boundaries, stability failures, and transport singularities.*

Axiom 4.5 (Composition and gluing). *Local comparison maps preserve identities and composition. On overlaps,*

$$T_{ii} = \text{id}, \quad T_{jk}T_{ij} = T_{ik}. \quad (10)$$

Assumption 4.6 (Regular classical metric module). *On regular finite classical statistical models, local distinguishability is represented by a continuous smooth tensor family invariant under sufficient Markov embeddings, with fixed positive normalization.*

Axiom 4.7 (Hard spectral resolution). *The resolved selector P is G -self-adjoint, idempotent, commutes with $L = G^{-1}F$, agrees with the threshold mask on every generalized eigenspace, and composes on independent direct sums.*

Definition 4.8 (Gate-admissible resolved subobjects). *Let $R_\tau = \text{im } P_\tau$. For fixed Γ , let $\text{Adm}_\Gamma(R_\tau)$ be the set of resolved subobjects satisfying all declared nuisance, stability, conditioning, coarse-graining, and transport gates.*

Axiom 4.9 (Admissibility soundness). *The reported sector belongs to the admissible family:*

$$W_{\text{obs|adm}} \in \text{Adm}_\Gamma(R_\tau). \quad (11)$$

Equivalently, it contains no direction rejected by any declared gate.

Axiom 4.10 (Admissibility completeness). *The reported sector is greatest among gate-admissible resolved subobjects:*

$$U \in \text{Adm}_\Gamma(R_\tau) \implies U \subseteq W_{\text{obs|adm}}. \quad (12)$$

Thus $W_{\text{obs|adm}}$ retains every direction accepted by the complete declared protocol.

Remark 4.11. *Soundness is a safety condition; completeness is a non-arbitrariness condition. Soundness alone permits arbitrary strict reductions. Completeness forbids discarding an accepted direction without adding a new declared gate.*

5 Characterization ladder

Theorem 5.1 (Quotient representation). *Under observational extensionality, every experiment-internal map $\Phi : \Theta \rightarrow Z$ factors uniquely through $Q_\mathcal{E}$. There is a unique $\tilde{\Phi} : Q_\mathcal{E} \rightarrow Z$ such that*

$$\Phi = \tilde{\Phi} \circ \pi_\mathcal{E}. \quad (13)$$

Proof. Define $\tilde{\Phi}([\theta]) = \Phi(\theta)$. Extensionality makes the definition representative-independent. Surjectivity of $\pi_\mathcal{E}$ gives uniqueness. $\square$

Corollary 5.2 (Canonical observable domain). *Any two realizations satisfying the quotient universal property are uniquely isomorphic by an isomorphism commuting with the quotient maps.*

Proof. This is the standard uniqueness argument for universal quotients: each candidate factors uniquely through the other, and uniqueness forces the composites to be identities. $\square$

Theorem 5.3 (Classical metric module). *Under the regular classical metric assumptions, the local distinguishability metric is Fisher–Rao up to the fixed normalization.*

Proof. This is the imported Čencov characterization module. The present paper does not reprove it and uses it only in its stated domain. $\square$

Theorem 5.4 (Unique hard resolved projector). *Let $F \succeq 0$, $G \succ 0$, and $\tau \notin \text{spec}(G^{-1}F)$. The unique operator satisfying hard spectral resolution is*

$$P_\tau = \mathbf{1}_{(\tau, \infty)}(G^{-1}F). \quad (14)$$

Proof. The operator $L = G^{-1}F$ is G -self-adjoint and therefore admits a G -orthonormal spectral decomposition. Commutation makes each eigenspace invariant. Idempotence and G -self-adjointness restrict the selector on each eigenspace to an orthogonal projection. Threshold consistency fixes that projection to identity above τ and zero at or below τ . Functional calculus gives the displayed projector and direct-sum composition. $\square$

Lemma 5.5 (Intersection realization of a finite linear protocol). *Suppose the fixed gates of Γ are represented by linear subspaces $A_1, \dots, A_k \subseteq R_\tau$. Then*

$$W_* = \bigcap_{a=1}^k A_a \quad (15)$$

is the unique greatest subspace satisfying every gate.

Proof. The intersection is a subspace contained in every A_a , hence it is sound. If U is contained in every A_a , then by the definition of intersection $U \subseteq W_*$. Hence W_* is greatest. If W' is another greatest element, then $W_* \subseteq W'$ and $W' \subseteq W_*$, so $W' = W_*$. $\square$

Theorem 5.6 (Maximal admissible-sector characterization). *In a finite-dimensional linear protocol fibre whose declared gates admit a greatest common resolved subobject, soundness and completeness characterize a unique final sector:*

$$W_{\text{obs|adm}} = \max \text{Adm}_\Gamma(R_\tau). \quad (16)$$

For linear subspace gates this becomes

$$W_{\text{obs|adm}} = R_\tau \cap A_N \cap A_S \cap A_C \cap A_T. \quad (17)$$

Proof. Existence follows from the assumed greatest admissible subobject, or from the intersection lemma in the linear case. If W and W' both satisfy soundness and completeness, completeness of W applied to W' gives $W' \subseteq W$, while completeness of W' applied to W gives $W \subseteq W'$. Thus $W = W'$. $\square$

Corollary 5.7 (No undeclared shrinkage). *Any strict subobject $U \subsetneq W_{\text{obs|adm}}$ violates admissibility completeness unless an additional gate excluding $W_{\text{obs|adm}} \setminus U$ is added to Γ .*

Corollary 5.8 (Conditional uniqueness of $W_{\text{obs|adm}}$). *The final admissible sector is unique conditional on the complete protocol but is not universal with respect to changes of protocol.*

Theorem 5.9 (Natural transport of fixed-protocol geometry). *Let $f : (\mathcal{E}, \Gamma) \rightarrow (\mathcal{E}', \Gamma')$ be an experiment isomorphism transporting the complete protocol. Then f induces an isomorphism of quotient domains, transports the Fisher tensor covariantly, conjugates the hard projector, and maps the maximal admissible sector onto the maximal admissible sector:*

$$f_* W_{\text{obs|adm}} = W'_{\text{obs|adm}}. \quad (18)$$

Proof. Quotient naturality follows from equality-of-law preservation. Metric covariance is part of the metric module. Spectral functional calculus is equivariant under conjugation. Since the complete gate family is transported, $f_*W_{\text{obs|adm}}$ is admissible in the target. For any target-admissible U' , its inverse image is source-admissible and hence lies in $W_{\text{obs|adm}}$; therefore $U' \subseteq f_*W_{\text{obs|adm}}$. Thus the image is target-maximal and equals the unique target sector. $\square$

Theorem 5.10 (Fixed-protocol observable-geometry representation). *Let $(\mathcal{E}, \Gamma)$ be a regular finite-dimensional classical declared admissible experiment. Assume:*

1. *observational extensionality and quotient universality;*
2. *the regular classical Fisher metric module with declared normalization;*
3. *a positive comparison metric G , a threshold τ separated from the generalized spectrum, and the hard spectral-resolution axioms;*
4. *a gate family admitting a greatest admissible resolved subobject;*
5. *admissibility soundness and completeness;*
6. *natural transport, identity, cocycle gluing, and independent-product composition.*

Then the observable geometry

$$\text{RG}(\mathcal{E}, \Gamma) = (Q_{\mathcal{E}}, g_F, P_{\tau}, W_{\text{obs|adm}}, \{T_{ij}\}, \Gamma) \quad (19)$$

is determined uniquely up to protocol-preserving natural isomorphism and the declared Fisher normalization.

Proof. The quotient representation theorem uniquely fixes $Q_{\mathcal{E}}$ up to unique compatible isomorphism. The imported metric characterization fixes g_F up to its declared normalization. The spectral projector theorem uniquely fixes P_{τ} . The maximal admissible-sector theorem uniquely fixes $W_{\text{obs|adm}}$. Naturality uniquely determines transported copies from these fixed local data, while identity and cocycle compatibility make direct and composite identifications agree on overlaps. Product composition excludes a distinct assignment on independent sums. Therefore two realizations of the same bundled input are connected componentwise by protocol-preserving isomorphisms. The quotient map and transported structural maps determine the compatible isomorphism; the componentwise naturality squares commute. Hence the realizations are naturally isomorphic. $\square$

Remark 5.11 (Exact force of established literature). *The quotient and metric layers inherit genuine established characterization theorems. The projector layer is a direct consequence of the spectral theorem and the fixed threshold mask. The admissible-sector layer is a greatest-subobject argument in a subspace lattice. Category theory then supplies the correct uniqueness language: universal or greatest constructions are unique only through structure-preserving maps. The full RG theorem is therefore modular; no cited classical theorem, by itself, states the assembled RG result.*

Corollary 5.12 (Fibrewise canonicity). *For each fixed admissibility protocol Γ , the RG assignment defines a canonical functor*

$$\text{RG}_{\Gamma} : \text{Exp}_{\Gamma} \longrightarrow \text{ObsGeom}_{\Gamma} \quad (20)$$

within the theorem's domain. There is no asserted canonical identification between fibres belonging to different protocols.

Corollary 5.13 (No canonical protocol-free section). *The representation theorem does not provide a map $\mathcal{E} \mapsto \Gamma$. Consequently, it does not define a canonical section of the projection from declared experiments to bare experiments.*

6 Finite fixed-protocol attack

6.1 Design

A fixed six-mode protocol was declared in a G -orthonormal information eigenbasis:

$$\lambda = (0.18, 0.42, 0.91, 1.37, 2.15, 3.80), \quad \tau = 1. \quad (21)$$

The nuisance, stability, conditioning, coarse-graining, and transport masks were frozen before enumeration. The canonical resolved and admissible masks were

$$P_\tau : (0, 0, 0, 1, 1, 1), \quad (22)$$

$$W_{\text{obs|adm}} : (0, 0, 0, 1, 0, 0). \quad (23)$$

Every binary hard spectral projector mask and every binary final-sector mask were paired, giving

$$2^6 \times 2^6 = 4096 \quad (24)$$

competitors.

6.2 Elimination ledger

Table 1: Survivors in the exhaustive fixed-protocol attack.

Constraint	Survivors
Initial hard spectral/sector pairs	4096
Fixed-threshold consistency	64
Gate soundness	2
Gate completeness/maximality	1
Transport naturality	1
Product composition	1

Exactly one pair survived, equal to the canonical construction. Soundness alone left two competitors: the canonical one-dimensional sector and its zero strict subspace. Completeness removed the zero sector. Thus the finite attack did not merely fail to find alternatives; it isolated an explicit independence witness for the new axiom.

A second attack generated 128 random oblique similarities of the canonical projection. None simultaneously satisfied idempotence, metric self-adjointness, and commutation with the information operator.

Proposition 6.1 (Finite battery result). *Within the enumerated six-mode hard-spectral class, the fixed protocol admits exactly one observable-sector assignment satisfying threshold consistency, gate soundness and completeness, natural transport, and product composition.*

Proof. Exhaustive enumeration checks all 4096 binary spectral/projected-sector pairs. The survivor ledger above is generated deterministically by the accompanying script. The unique surviving masks coincide with the analytic construction. $\square$

Remark 6.2. *The computation is not a proof of the representation theorem. The theorem follows from the modular arguments above. The battery instead checks implementation, exhibits the necessity of completeness in the finite model, and searches the full stated binary competitor class for overlooked alternatives.*

7 Axiom roles and independence

Proposition 7.1 (Soundness does not imply completeness). *There exist fixed protocols for which more than one sound sector exists.*

Proof. If the canonical maximal sector is nonzero, both it and the zero subspace satisfy containment in every gate. Only the maximal sector satisfies completeness. $\square$

Proposition 7.2 (Completeness does not select a protocol). *Admissibility completeness is conditional on Γ and does not derive G , τ , N , S , C , or T from $\mathcal{E}$.*

Proof. Different protocols define different admissible families $\text{Adm}_\Gamma(R_\tau)$ and therefore may have different greatest elements. Completeness selects the greatest object within a declared family; it does not select the family. $\square$

Proposition 7.3 (Threshold consistency is independent of hard projection). *Idempotence, metric self-adjointness, and commutation with L admit every orthogonal spectral subprojection. Threshold consistency is required to select the specified upper spectral set.*

Proof. In a spectral basis, any binary choice on eigenspaces defines a commuting orthogonal projector. Only the threshold mask fixes which eigenspaces are retained. $\square$

8 Interpretation

The representation theorem identifies RG with neither a bare quotient nor a universal final sector. The quotient is universal at the domain layer. The final sector is maximal only inside a fixed protocol fibre. The complete logical order is

$$(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow g_F \longrightarrow P_\tau \longrightarrow W_{\text{obs|adm}} \longrightarrow \{T_{ij}\}. \quad (25)$$

The word “canonical” must therefore be read fibrewise. A protocol is to RG what a metric is to Riemannian geometry: declared structure whose consequences are characterized. This is weaker than protocol-free uniqueness and stronger than presenting an arbitrary construction. It says that no additional undeclared choice remains after the complete admissibility contract and the RG axioms are fixed.

The theorem schema is modular. A future quantum version could replace the classical Fisher module with a characterized quantum distinguishability module while retaining quotient extensionality, hard or graded spectral resolution, admissibility maximality, and natural transport, provided the appropriate categories and subobject meets exist. No such extension is proved here.

9 Non-claims and falsification criteria

Non-claim 9.1 (No protocol-free uniqueness). *The paper does not assert that $\mathcal{E}$ determines Γ , or that all observation geometries are RG.*

Non-claim 9.2 (No universality of $W_{\text{obs|adm}}$). *The universal object is the equality-of-law quotient for experiment-internal maps. The final admissible sector is a maximal subobject relative to a fixed protocol, not an initial or terminal object across all protocols.*

Non-claim 9.3 (No unrestricted Fisher theorem). *Fisher–Rao is imported only in the regular finite classical domain under the stated invariance and normalization assumptions. Singular, infinite-dimensional, non-dominated, and quantum experiments require separate metric modules.*

Non-claim 9.4 (No automatic global uniqueness). *Atlas uniqueness is relative to declared transition data satisfying identity and cocycle laws. Different undeclared connections or holonomy classes are different protocols, not rival realizations in one fibre.*

Non-claim 9.5 (No physical ontology). *The quotient expresses operational indistinguishability. It neither denies latent representatives nor identifies the observable geometry with spacetime, quantum state reality, or any physical ontology.*

Non-claim 9.6 (No infinite-dimensional maximality without hypotheses). *In general categories or topological vector spaces, arbitrary gate intersections may fail to belong to the intended regularity class or may require closure. The finite linear greatest-subobject proof does not automatically extend.*

Falsification and revision conditions. The stated characterization fails or requires revision if:

1. an experiment-internal quantity varies inside an equality-of-law class;
2. a metric satisfying the stated classical module is not proportional to Fisher–Rao;
3. two distinct hard projectors satisfy all fixed (F, G, τ) selector axioms off the spectrum;
4. two distinct greatest admissible subobjects exist in the same fixed gate family;
5. covariant transport changes the characterized object beyond natural isomorphism;
6. direct and composite overlap transports disagree;
7. a claimed stable sector has no positive margin from the declared discriminant.

10 Open problems

1. Prove existence of greatest admissible subobjects for nonlinear, stratified, and sheaf-valued protocol categories.
2. Determine the correct closure operation for infinite-dimensional gate intersections and establish perturbation-stable maximality.
3. Extend the representation theorem to singular and rank-changing experiments.
4. Formulate a quantum experiment category and replace the classical metric module by an explicitly characterized monotone quantum module.
5. Separate protocol-preserving natural equivalence from equivalence under protocol refinement or Blackwell comparison.
6. Determine when decision, resource, or physical axioms select a subfamily of protocols; this is external to the present fibrewise theorem.

11 Conclusion

The correct classical identification is now precise. Resolution Geometry is the canonical observable geometry of a declared admissible experiment, not the unique protocol-free geometry of observation. Quotient universality fixes the observable domain. The classical metric characterization fixes Fisher–Rao distinguishability in its domain. Spectral theory fixes the hard resolved projector. The newly explicit completeness axiom fixes the final sector as the greatest resolved

subobject satisfying every declared gate. Naturality and cocycle laws fix transport and gluing relative to the contract.

Thus the defensible uniqueness statement is fibrewise:

$$(\mathcal{E}, \Gamma) \Longrightarrow \text{RG}(\mathcal{E}, \Gamma) \quad \text{up to protocol-preserving natural isomorphism.} \quad (26)$$

The finite battery supports this statement, exposes completeness as necessary, and finds no competing hard-spectral realization in the exhaustive class. Protocol selection remains outside the theorem by design.

Reproducibility

The deterministic supporting artifacts are:

- `rg_fixed_protocol_uniqueness_attack.py`;
- `rg_fixed_protocol_uniqueness_attack.json`.

The seed is 20260805. The exhaustive binary enumeration itself is seed-independent; the seed controls the supplementary oblique-projector attack.

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A Battery pseudocode

```
fix information spectrum, threshold, and all protocol gates
canonical_P = spectral_mask(information > threshold)
canonical_W = canonical_P AND all_gate_masks

for P in all binary spectral projectors:
  for W in all binary sector masks:
    require P == canonical_P                # threshold consistency
    require W subset_of (P AND all_gates)    # soundness
    require W == (P AND all_gates)          # completeness
    require covariance under transported protocol
    require direct-sum composition
    retain(P,W)

assert exactly one retained pair
assert retained pair == (canonical_P, canonical_W)
```

B Machine-readable result summary

all competitors:	4096
survive threshold consistency:	64
survive gate soundness:	2
survive gate completeness:	1
survive natural transport:	1
survive product composition:	1
random oblique projectors tested:	128
oblique survivors of all projector rules:	0